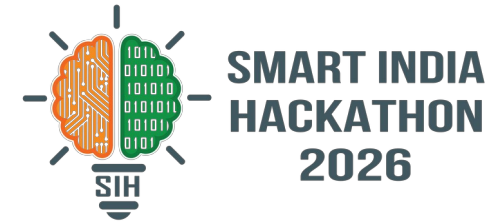
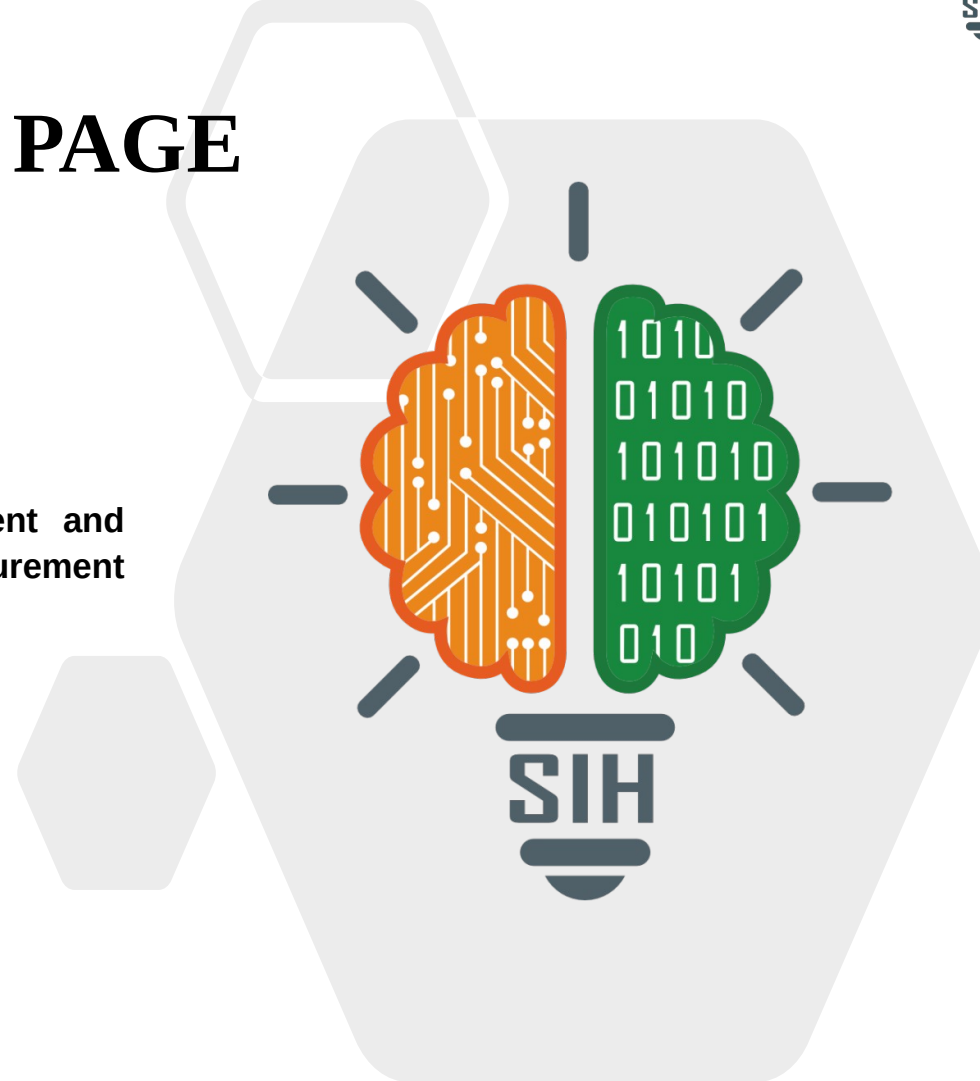


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – SIH26031**
- **Problem Statement Title-** Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- [Team ID]**
- **Team Name- StrataCadastre_KP**



PROPOSED SOLUTION

EkMaap ("one measure"): one photo of the sample → every onion measured and labelled → a verifiable quality report.

The gap

An onion lot's grade decides whether it is bought and at which rate, yet it is judged by eye at each procurement centre. Two officers can grade the same lot differently — the source of disputes.

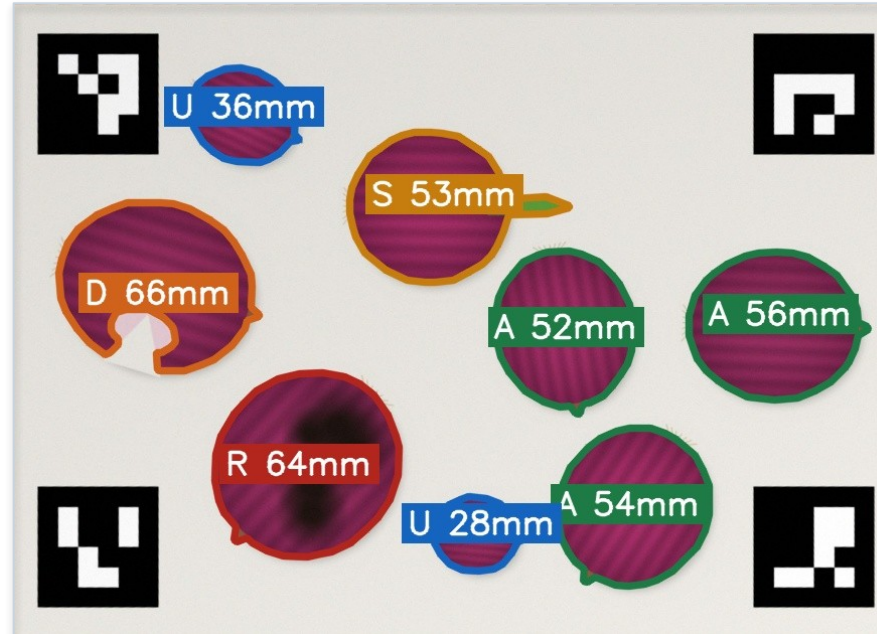
Why it matters

Norms change mid-season: in 2026 NAFED's minimum size moved to 45 mm in June and URS purchases stopped on 30 July. Manual grading keeps no record of WHY a lot got its grade.

Our solution

A phone app + server. The officer photographs the sample on a printed ArUco mat. AI outlines every onion, measures it in mm, flags damaged, rotten, sprouted and undersized bulbs, applies the current rule set and issues a fingerprinted report in seconds.

ONE PHOTO, EVERY ONION MEASURED AND LABELLED



Prototype output — A Grade A · U undersized · S sprouted · D damaged · R rotten

OUTPUT PER LOT



+ digital report



WHAT CHANGES

Today (manual)	With EkMaap
Grade judged by eye	Grade computed from the photo
Size judged visually	Size in mm, tilt-corrected
No record of why	Every onion's result stored
New norms applied unevenly	One versioned rule set
Dispute is word vs word	Photo + fingerprint as evidence
Unsure cases decided silently	Unsure onions go to the officer

SIX LABELS PER ONION



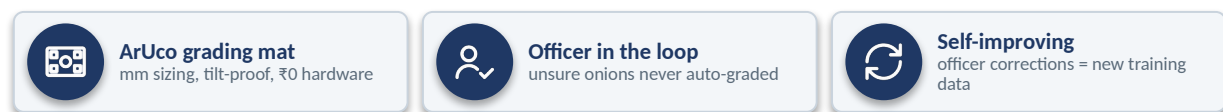
PROCESSING PIPELINE



SYSTEM ARCHITECTURE — LAYERED STACK

 FRONTEND	Officer web app (PWA) · phone camera · review screen · no install
 BACKEND	FastAPI · REST API with role-based access (officer / supervisor / labeller)
 DATABASE	PostgreSQL · append-only analyses, reviews, reports, audit log
 VISION	OpenCV ArUco homography · scikit-image watershed · moment-based sizing
 AI / ML	Trainable pixel segmenter + condition classifier (scikit-learn) · YOLO-seg on GPU
 RULES	Versioned rule sets: size limits, condition → grade; editable per circular
 REPORTS	ReportLab PDF · SHA-256 fingerprint · QR · public verify endpoint

KEY INNOVATIONS



SEE IT RUNNING



WORKING PROTOTYPE

[Click here to see the working prototype.](#)



GITHUB REPO

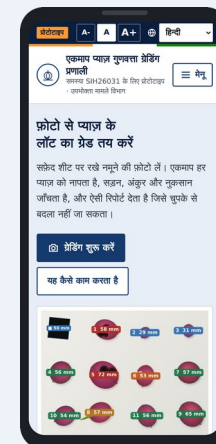
[Click here to see the GitHub repository.](#)



DEMO VIDEO

[Click here to see the demo video.](#)

Built: calibration, segmentation, sizing, classifier, rule engine, fingerprinted PDF reports, dispute flow, dataset + model registry — REST API over PostgreSQL.



Prototype in Hindi

USER JOURNEY



FEASIBILITY AND VIABILITY



ECONOMIC FEASIBILITY



Uses phones already in hand

Web app in the officer's phone browser. The only extra item is a printed A4 mat.



No licence cost

Open-source stack (OpenCV, scikit-learn, FastAPI, PostgreSQL), self-hosted in Docker.



Fewer re-grades

Stored photo + report settle a dispute without re-grading the whole lot.



TECHNICAL FEASIBILITY



Working prototype

Full chain runs end to end: photo → mm sizing → condition → grade → verified PDF.



Trains on a laptop

Segmenter and classifier train on CPU in minutes; GPU model optional at scale.



Proven building blocks

Marker-based metrology and image classification are mature computer-vision methods.



OPERATIONAL FEASIBILITY



Fits today's process

Officer still draws the sample; the app replaces judging by eye, not the officer.



Norms change without code

New limits = new rule-set version, applied to every centre at once.



Runs in NIC / state data centres

Self-hosted Docker stack; photos never leave government servers.



SOCIAL FEASIBILITY



Farmer sees the evidence

Same annotated photo and per-onion result for farmer and officer.



Same lot, same grade

Any centre, same rule version → same answer from the same photo.



Local languages

Interface in Hindi, Marathi, Gujarati, Tamil, Telugu, Kannada and English.

KEY RISKS AND HOW WE OVERCOME THEM



Varied light and backgrounds

Matte printed mat + capture guide; models retrained on photos from each new centre.



Onions touching or clustered

Learned boundary class + watershed splitting; officer confirms any flagged onion.



Internal rot is invisible

Report lists surface grade; the officer's cut test stays part of the procedure.



Grade norms keep changing

Limits live in versioned rule sets, updated by a supervisor per official circular.

IMPLEMENTATION ROADMAP



Phase 1 • Data

Label mandi onion photos; ruler-measure a subset



Phase 2 • Train

Train segmenter + classifier; test on held-out photos



Phase 3 • Pilot

One procurement centre, officers in the loop



Phase 4 • Scale

Multi-centre rollout; retrain from officer corrections

DETAILED FINAL REPORT



[Click here to see the detailed report.](#)

Architecture, database, API and training plan in full.

IMPACT AND BENEFITS



2 lakh t

onion procurement target for 2026 (NAFED + NCCF, 1 lakh t each)³



₹2,335/qrtl

Grade A procurement price from 30 July 2026²



45 mm

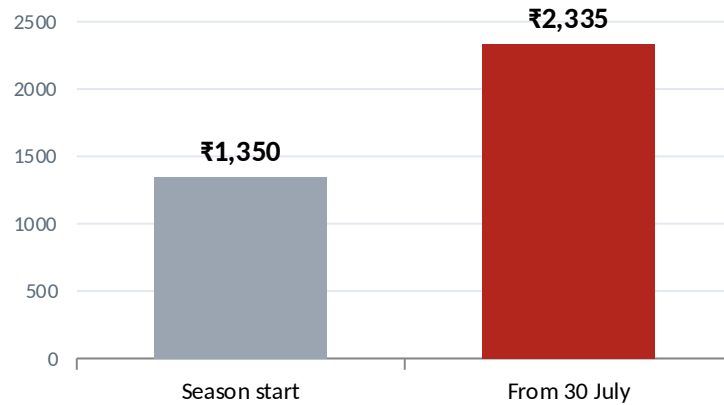
minimum size NAFED accepted after the June 2026 relaxation (was ~55 mm)¹



7th

procurement price hike of the season, by 30 July 2026²

GRADE A PROCUREMENT PRICE, 2026 (₹/QUINTAL)



Source: Free Press Journal, 31 Jul 2026²

GRADING RULES CHANGED MID-SEASON (2026)



June 2026 Norms relaxed¹

Min. size ~55 → 45 mm; one peeled layer, up to 30% blackening, 40% spots, 10% sunburn accepted.



30 July 2026 URS stopped²

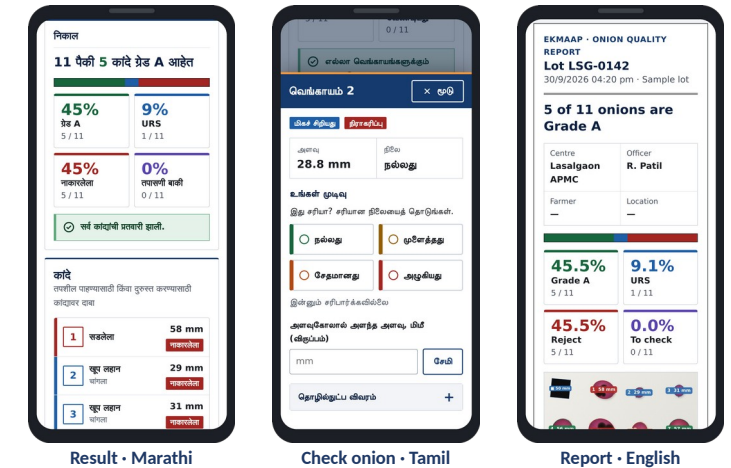
Only Grade A bought from this date; Grade A price raised to ₹2,335/qrtl.



Next change One click

In EkMaap: a new rule-set version, live at every centre the same day.

PROTOTYPE — OFFICER SCREENS



Officer confirms flagged onions, then issues the report.

IMPACT ON THE TARGET AUDIENCE



Farmers

See exactly why their lot got its grade; contest with evidence, not argument.



Procurement officers

Grade backed by photo + rule version; less personal blame in disputes.



NAFED / NCCF / DoCA

Comparable quality data across centres; new norms rolled out as one update.



Consumers

Cleaner intake into the buffer stock released to stabilise onion prices.



SOCIAL

Fewer disputes at the centre; trust built on shared evidence



ECONOMIC

Right price for the right grade; fewer costly re-grades



GOVERNANCE

Auditable, versioned, tamper-evident grading at every centre

RESEARCH AND REFERENCES



GRADING STANDARDS & POLICY



SIH26031 problem statement

DoCA, Ministry of Consumer Affairs — expected solution

sih.gov.in



Fruits & Vegetables Grading and Marking Rules, 2004

AGMARK onion standard (Schedule XIX)

agritech.tnau.ac.in/amis/pdf/ONION_AGMARK.pdf



Price Stabilisation Fund

Scheme under which the onion buffer is procured

consumeraffairs.nic.in



NAFED • NCCF

Central procurement agencies for the onion buffer

nafed-india.com • nccf-india.com



2026 PROCUREMENT FACTS (NEWS)



¹ NAFED relaxes onion procurement norms

Free Press Journal • 4 Jun 2026

freepressjournal.in/pune



² Grade A rate ₹2,335/qtl; URS stopped 30 July

Free Press Journal • 31 Jul 2026

freepressjournal.in/pune



³ Early procurement, 2 lakh t target

Free Press Journal • 8 Apr 2026

freepressjournal.in/pune



Size range eased to 35–70 mm

ETV Bharat • 7 Jun 2026

etvbharat.com



TECHNOLOGY STACK



OpenCV ArUco

Marker detection, homography, rectification

docs.opencv.org



scikit-learn • scikit-image

Pixel segmenter, classifier, watershed

scikit-learn.org • scikit-image.org



FastAPI • PostgreSQL

REST API and audit-ready database

fastapi.tiangolo.com • postgresql.org



ReportLab • CVAT

PDF reports with QR • labelling tool

reportlab.com • cvat.ai

FIELD VALIDATION PLAN



1. Mandi sample study

good and defective onions photographed on the mat and measured by ruler — the labelled training and test set.



2. Procurement-centre observation

how officers draw and grade samples today, and which written norms they apply.



3. Grader-agreement study

six graders vs the app on the same onions — measures the subjectivity the PS describes. [Click here to see the research report.](#)